rigorous oversight, robust cyber and electronic-warfare protections, and international dialogue on acceptable use.

Figure 9.1 illustrates a heterogeneous military drone swarm operating under a coordinated autonomous mission framework. A large, long-endurance mothership UAV functions as the central airborne command platform, distributing tasks and mission objectives to multiple classes of smaller unmanned aerial vehicles. Surrounding the mothership are sensor drones, equipped with cameras and reconnaissance payloads used for terrain mapping, target identification, and situational awareness. Also present are electronic warfare drones, designed to conduct jamming, deception, and communication disruption against opposing forces, and loitering munition drones, which serve as precision strike units capable of engaging high-value targets. Communication between all drones occurs through a distributed mesh network, allowing realtime coordination, dynamic role reassignment, and resilience even if individual drones are disabled. The figure demonstrates how swarm intelligence enables drones to operate collectively as a flexible, adaptive, and mission-cooperative aerial combat system, rather than as isolated platforms.